

SETU-Shield

A privacy-preserving safety net for every UPI payment

Track: Real-Time Payments | {{TEAM_NAME}} | {{COLLEGE}}

The money is gone in minutes

- Victim is talked into paying a collect request or QR
- Funds hop through mule accounts across banks within minutes
- By the time a complaint is filed, the trail has gone cold

Every bank sees one hop

- Each bank sees only the edges touching its own customers
- No bank can see the whole ring, and raw data cannot be pooled (privacy, DPDP Act)
- The user still gets a generic "Are you sure?" popup that they ignore

Our idea: three layers

1 Federate

Banks train one shared mule model. Only model updates travel, never customer data.

2 Score

Payee risk is scored before money moves, with a reason a person can read, in their language.

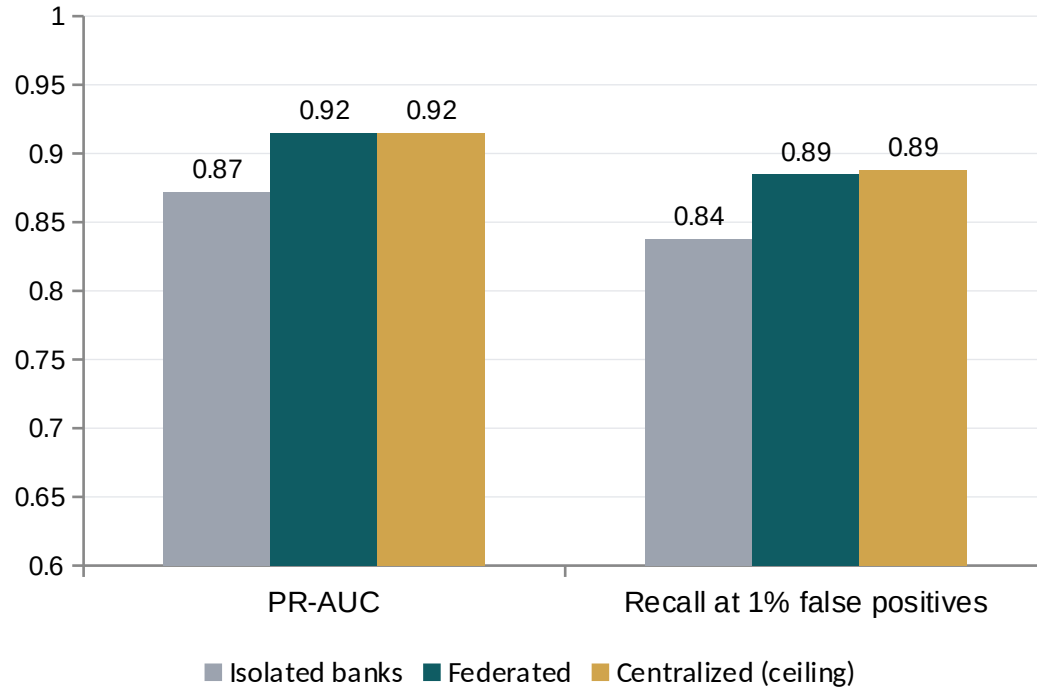
3 Guardian

For opted-in users, a risky payment is held and a family member approves or rejects it.

Data stays in the bank. Only updates travel.

- Each bank trains locally on its own partial transaction graph
- Weighted FedAvg combines the updates into one global model
- Differential privacy and secure aggregation are designed in but not yet measured or built to production grade

Does it work? On synthetic data, yes



Federated matches the pooled-data ceiling and beats banks working alone.

3 simulated banks, 5 seeds, global test set. Not a real-world benchmark.

A safety net for people, not just banks

- Calm reason in English, Hindi, Tamil or Telugu, with voice
- Held payment sends a signed, single-use link to a trusted contact
- No reply in 10 minutes cancels the payment (safe default)
- If the scorer is down, payments continue (fail open)

What we won't pretend

- Data is simulated; no real NPCI, bank or customer data was used
- Serving today uses a rule-based policy; the federated model is validated offline
- Secure aggregation, DP accounting and poisoning defences are future work
- Fairness across user groups and translation review still needed

Where this goes

- Wire the federated model into the scoring API
- Shadow-mode pilot with one bank, then an advisory signal beside existing fraud systems
- **Ask: mentorship, sandbox access, and a pilot introduction**